Supplementary Material for "Battery Dispatch Optimization Considering Cell Voltage Profiles: A Piecewise Input Convex Neural Network Approach"

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I. Proof of Theorem 1

With given $W^t_{i,j,p}$, $V^t_{i,j,p}$, and $b^t_{i,j,p}$ in (1a) for the trained PICNN, the bilinear term can be relaxed using the McCormick linearization method, as each ReLU activation function is inherently continuous:

$$z^{t+1}_{i,j,p} \leqslant \sigma((x^t_i)^{\mathrm{T}} W^t_{i,0,p} + b^t_{i,0,p}) + (1-\alpha^t_{i,p})M, \quad \text{if } j=1, \tag{I-1a}$$

$$z^{t+1}_{i,j,p} \geqslant \sigma((x^t_i)^{\mathrm{T}} W^t_{i,0,p} + b^t_{i,0,p}) + (\alpha^t_{i,p}-1)M, \quad \text{if } j=1, \tag{I-1b}$$

$$\begin{aligned} z^{t+1}_{i,j,p} \leqslant{}& \sigma((z^{t+1}_{i,j-1,p})^{\mathrm{T}} W^t_{i,j-1,p} + (x^t_i)^{\mathrm{T}} V^t_{i,j-1,p} + b^t_{i,j-1,p}), \\ &+(1-\alpha^t_{i,p})M, \qquad \text{if } j=2,\ldots,N_I-1, \end{aligned} \tag{I-1c}$$

$$\begin{aligned} z^{t+1}_{i,j,p} \geqslant{}& \sigma((z^{t+1}_{i,j-1,p})^{\mathrm{T}} W^t_{i,j-1,p} + (x^t_i)^{\mathrm{T}} V^t_{i,j-1,p} + b^t_{i,j-1,p}), \\ &+(\alpha^t_{i,p}-1)M, \qquad \text{if } j=2,\ldots,N_I-1, \end{aligned} \tag{I-1d}$$

$$z^{t+1}_{i,j,p} = (z^{t+1}_{i,j-1,p})^{\mathrm{T}} W^t_{i,j-1,p} + (x^t_i)^{\mathrm{T}} V^t_{i,j-1,p} + b^t_{i,j-1,p}, \ \text{if } j = N_I, \tag{I-1e}$$

where M is a large positive scalar. Each output $z^{t+1}_{i,j,p}$ is convex with respect to the input vector x^t_i for the p-th range of variable x^t_i, i.e., $\alpha^t_{i,p}=1$, if all linear-term coefficients are nonnegative, and the activation function $\sigma(\cdot) = \max(\cdot, 0)$ is convex and non-decreasing. Then, (A-1a) and (A-1c) naturally hold. After neglecting (A-1a) and (A-1c), (A-1a)–(A-1e) simplify as

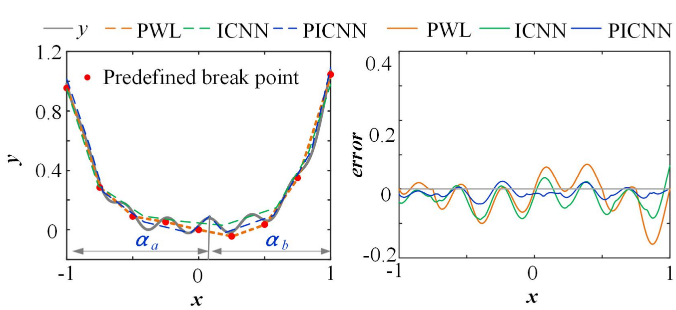
$$\min_{x^t_i, z^{t+1}_{i,j,p}, \alpha^t_p} \sum_{p\in\mathcal{P}} z^{t+1}_{i,N_I,p}, \tag{I-2a}$$

$$z^{t+1}_{i,j,p} \geqslant (x^t_i)^{\mathrm{T}} W^t_{i,0,p} + b^t_{i,0,p} + (\alpha^t_p - 1)M, \quad \text{if } j=1, \tag{I-2b}$$

$$\begin{aligned} z^{t+1}_{i,j,p} \geqslant{}& (z^{t+1}_{i,j-1,p})^{\mathrm{T}} W^t_{i,j-1,p} + (x^t_i)^{\mathrm{T}} V^t_{i,j-1,p} + b^t_{i,j-1,p}, \\ &+(\alpha^t_p - 1)M, \qquad \text{if } j=2,\ldots,N_I-1, \end{aligned} \tag{I-2c}$$

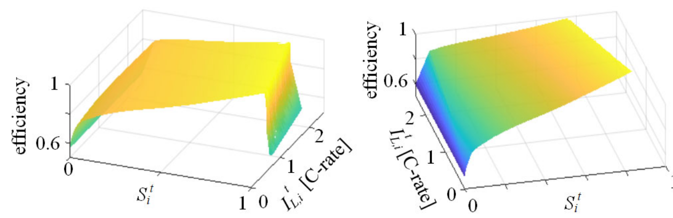
$$z^{t+1}_{i,j,p} = (z^{t+1}_{i,j-1,p})^{\mathrm{T}} W^t_{i,j-1,p} + (x^t_i)^{\mathrm{T}} V^t_{i,j-1,p} + b^t_{i,j-1,p}, \ \text{if } j = N_I. \tag{I-2d}$$

Clearly, this model (A-2a)–(A-2d) is the same with the MILP-based model (2a)–(2d), which proves the Theorem. ∎



(a) (b)

Fig. 1. Approximation comparison among PWL, ICNN and PICNN.



(a) (b)

Fig. 2. Efficiency coefficients: (a) $\eta_{c,i}$ under CCCV charging; and (b) $\eta_{d,i}$ under CC discharging.

II. Comparing PWL-, ICNN-, and PICNN-based Approximations

We compare the three methods via a simple example $y = x^4 + 0.05\sin(20x)$ over $x \in [-1, 1]$, as shown in Fig. 1(a). For the traditional PWL method, the linear segments are manually predefined with equal length, using a fixed number of 8 segments. Both ICNN and PICNN have 4 layers (with dimensions 16, 8, 4, and 1), trained with an initial learning rate of 0.02, decaying by 0.005 over 1200 iterations. In the PICNN, two partitions of $x \leqslant 0.083$ and $x > 0.083$ are indicated by binary indicators α_a and α_b. The ICNN and PICNN are trained using 1024 sample data points. As shown in Fig. 1(b), the maximum absolute error of the traditional PWL method, the ICNN method, and the proposed PICNN method are 0.16, 0.09, and 0.06, respectively. This comparison demonstrates that the proposed PICNN method achieves greater approximation accuracy than the two existing methods, with the lowest maximum absolute error, owing to the use convex ranges indicated by binary variables informed by domain knowledge.

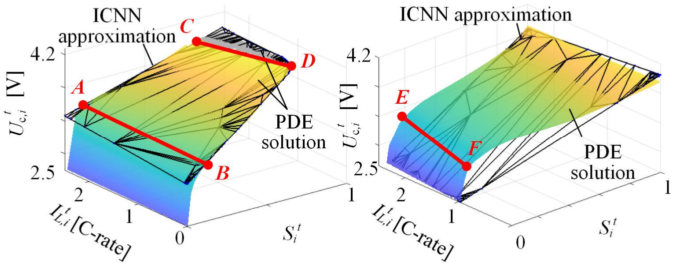
III. Charging Current Dynamics under CCCV Mode

The classical battery SoC equation is expressed as [5]

$$\Delta S^t_i = \eta_{\mathrm{c},i} p^t_{\mathrm{c},i} \Delta t + \eta_{\mathrm{d},i} p^t_{\mathrm{d},i} \Delta t, \tag{III-3}$$

where $p^t_{\mathrm{c},i}$ and $p^t_{\mathrm{d},i}$ respectively denote charging and discharging power of UES i's battery with fixed charging and discharging efficiency coefficients $\eta_{\mathrm{c},i}$ and $\eta_{\mathrm{d},i}$.

For $\Delta t = 1$ hour, we can estimate $\eta_{c,i}$ and $\eta_{d,i}$ via COMSOL simulation under CCCV charging and CC discharging cycles. As shown in Fig. 2(a) and (b), charging and discharging efficiency coefficients $\eta_{c,i}$ and $\eta_{d,i}$ are not fixed but rather vary in the ranges $[0.6, 0.98]$ and $[0.6, 0.99]$ depending on different values of S^t_i and $I^t_{\mathrm{L},i}$. Thus the classical battery



(a) (b)

Fig. 3. Voltage surfaces of a battery cell: (a) charging in CCCV mode; and (b) discharging in CC mode.

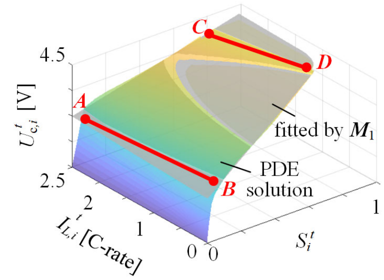


Fig. 4. Fitting the cell voltage profile using M_1 [6].

SoC equation with fixed efficiency coefficients lead to the inaccurate modelling of SoC variations. In contrast, the ICNN-based ΔS_i^t learned from PDE-based simulation data yields more accurate SoC approximations and further eliminates the need for charging and discharging power decision variables in the BDO problem.

IV. Comparing PICNN and ICNN Approaches

We compare the proposed multiple ICNN-based approximation with the conventional ICNN approach [7]. Figs. 3(a) and (b) respectively plot the voltage surfaces of charging a battery cell in CCCV mode and discharging in CC mode using the conventional ICNN approach. The conventional ICNN consists of $N_I = 4$ layers with dimensions of 16, 8, 4, and 1, respectively, with initial learning rate 0.02 decaying by 0.005 over 1200 iterations. It is trained using a total of 2,500 sample data points generated by COMSOL simulations.

We see that the approximation errors in Fig. 3 are clearly larger than the PICNN-based approach shown in Fig. 2 in the main manuscript, as the ICNN-based approximations do not explicitly account for different operating regions. In contrast, employing multiple ICNN-based approximations for different SoC ranges achieves greater accuracy: the maximum error between the PDE-based solution and the proposed multi-ICNN approach is less than 2%, compared to 11% in Fig. 3(a) and 67% in Fig. 3(b) for single ICNN-based approximations.

V. Derivation of Second-order Conic Constraint

Let $\mathcal{F} = (z_{i,N_I,p}^{t+1})^2$, and then the epigraph constraint is expressed as

$$\text{epi } \mathcal{F} := \{(\gamma_{i,p}^{t+1}, z_{i,N_I,p}^{t+1}) | \gamma_{i,p}^{t+1} \geqslant \mathcal{F}(z_{i,N_I,p}^{t+1})\}. \tag{V-4}$$

This epigraph constraint can be derived as a second-order conic constraint. Define $f := (\gamma_{i,p}^{t+1} + 1)^2$ and $g := (\gamma_{i,p}^{t+1} - 1)^2 + (2z_{i,N_I,p}^{t+1})^2$. Then

$$f - g = 4\gamma_{i,p}^{t+1} - 4(z_{i,N_I,p}^{t+1})^2 \geqslant 0, \tag{V-5}$$

where the inequality holds because $\gamma_{i,p}^{t+1} \geqslant (z_{i,N_I,p}^{t+1})^2$. Rearrangement of (V-5) yields

$$(\gamma_{i,p}^{t+1} + 1)^2 \geqslant (\gamma_{i,p}^{t+1} - 1)^2 + (2z_{i,N_I,p}^{t+1})^2, \tag{V-6}$$

which is equivalent to the standard second-order conic constraint

$$\left\|\gamma_{i,p}^{t+1} + 1\right\|_2^2 \geqslant \left\|[\gamma_{i,p}^{t+1} - 1,\ 2z_{i,N_I,p}^{t+1}]^{\mathrm{T}}\right\|_2^2. \tag{V-7}$$

VI. Battery Parameter Values

We set the total NMC battery capacity $B_i = 200\,\text{Ah}$ for each unmanned electric ships (UES). The cell current $I_{\mathrm{L},i}^t \in [0, \overline{I}_{\mathrm{L},i}]$ with $\overline{I}_{\mathrm{L},i} = 2.5\,\text{C-rate}$. Other parameters of the NMC battery cell for electrochemical kinetics in COMSOL simulation are given in Table I.

The cell voltage profile constraint in M_1 [6] is formulated as

$$(U_{\mathrm{c},i}^t)^2 \leqslant \overline{U}_{\mathrm{c}} U_{\mathrm{c},i}^t - R_i p_{\mathrm{c},i}^t. \tag{VI-8}$$

As indicated in Fig. 4, the grey hyperplane represents the fixed cell voltage profile, which exhibits significant approximation error for the PDE solution by COMSOL especially when S_i^t approaches 0 or 1. Using M_1, the fixed resistance R_i for a battery cell can be fitted as 9.6 p.u.

TABLE I
Parameters for electrochemical kinetics

Parameter	Negative electrode	Separator	Positive electrode
electrode plate area (m^2)	0.163	0.163	0.163
electrode thickness (m)	$78 \cdot 10^{-6}$	$20 \cdot 10^{-6}$	$45 \cdot 10^{-6}$
Li^+ diffusion coefficient (m^2/s)	$3.9 \cdot 10^{-5}$	-	$1.8 \cdot 10^{-8}$
active electrode volume fraction (%)	0.6	-	0.6
electrolyte phase volume fraction (%)	0.3	-	0.3
max solid phase concentration (mol/m^3)	31507	-	49000
particle radius (m)	$6 \cdot 10^{-6}$	-	$5 \cdot 10^{-6}$
reaction rate efficiency (A/m^2)	$9.77 \cdot 10^{-2}$	-	$1.19 \cdot 10^{-2}$
exchange current density of side reaction (A/m^2)	10	-	10
initial electrolyte concentration (mol/m^3)	$1.25 \cdot 10^4$	$1.25 \cdot 10^4$	$1.25 \cdot 10^4$
Binder volume fraction (%)	0.1	-	0.1
Separator volume fraction (%)	-	0.4	-